WORKED SOLUTION

STEP III 1996, Question 3

Pure • Half-Angle Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q3

The key idea

An integrand built from $1/(\text{constant} + \text{constant} \cdot \cos x)$ is the classic case for the half-angle substitution $t = \tan\frac{1}{2}x$, which turns any rational function of $\sin x$ and $\cos x$ into a rational function of t. Here it produces a standard $\int \mathrm{d}t/(k^2 + t^2)$, giving an inverse tangent. The two "hence" results are then just a matter of rescaling each integrand into the form $1/(1 - a\cos x)$ and choosing a and θ.

Solution

The general integral

Put $t = \tan\frac{1}{2}x$. As x runs from 0 to θ, t runs from 0 to $T = \tan\frac{1}{2}\theta$, which is finite and positive because $0 < \frac{1}{2}\theta < \frac{1}{2}\pi$. The standard half-angle formulae give

$$\cos x = \frac{1 - t^2}{1 + t^2}, \qquad \mathrm{d}x = \frac{2\,\mathrm{d}t}{1 + t^2},$$

the second because $\mathrm{d}t/\mathrm{d}x = \frac{1}{2}\sec^2\frac{1}{2}x = \frac{1}{2}(1 + t^2)$. Then

$$1 - a\cos x = \frac{(1 + t^2) - a(1 - t^2)}{1 + t^2} = \frac{(1 - a) + (1 + a)t^2}{1 + t^2},$$

so the factors of $1 + t^2$ cancel and

$$\int_0^\theta \frac{1}{1 - a\cos x}\,\mathrm{d}x = \int_0^T \frac{2}{(1 - a) + (1 + a)t^2}\,\mathrm{d}t = \frac{2}{1 + a}\int_0^T \frac{1}{k^2 + t^2}\,\mathrm{d}t, \qquad k^2 = \frac{1 - a}{1 + a}.$$

Since $-1 < a < 1$, both $1 - a$ and $1 + a$ are positive, so $k = \sqrt{(1 - a)/(1 + a)}$ is a genuine positive real number. (This is exactly where the condition on a is used: if $|a| \geq 1$ the denominator could vanish and the integral would be improper or the answer a logarithm.) Using $\int \mathrm{d}t/(k^2 + t^2) = \frac{1}{k}\tan^{-1}(t/k)$,

$$\int_0^\theta \frac{1}{1 - a\cos x}\,\mathrm{d}x = \frac{2}{(1 + a)k}\tan^{-1}\frac{T}{k}.$$

Finally $(1+a)k = \sqrt{(1+a)^2}\sqrt{\frac{1-a}{1+a}} = \sqrt{1-a^2}$, so

$$\boxed{\int_0^\theta \frac{1}{1-a\cos x}\,\mathrm{d}x = \frac{2}{\sqrt{1-a^2}}\tan^{-1}\left(\sqrt{\frac{1+a}{1-a}}\tan\tfrac{1}{2}\theta\right).}$$

Ask yourself Why do we need $\theta < \pi$?

The substitution $t = \tan\frac{1}{2}x$ breaks down at $x = \pi$, where t would be infinite. Keeping $\theta < \pi$ means the whole range $[0,\theta]$ maps to the finite interval $[0, \tan\frac{1}{2}\theta]$, and the formula needs no special care. (Letting $\theta \to \pi$ gives $\pi/\sqrt{1-a^2}$, since $\tan^{-1}$ tends to $\frac{1}{2}\pi$.)

The first "hence"

Take out a factor of 2:

$$\frac{1}{2-a\cos x} = \frac{1}{2}\cdot\frac{1}{1-\frac{1}{2}a\cos x}.$$

For the integral to be finite we need $-2 < a < 2$, and then $b = \frac{1}{2}a$ satisfies $-1 < b < 1$, so the general result applies with a replaced by b and $\theta = \frac{1}{2}\pi$, where $\tan\frac{1}{4}\pi = 1$:

$$\begin{aligned}\int_0^{\frac{1}{2}\pi}\frac{1}{2-a\cos x}\,\mathrm{d}x &= \frac{1}{2}\cdot\frac{2}{\sqrt{1-\frac{1}{4}a^2}}\tan^{-1}\sqrt{\frac{1+\frac{1}{2}a}{1-\frac{1}{2}a}}\\ &= \frac{2}{\sqrt{4-a^2}}\tan^{-1}\sqrt{\frac{2+a}{2-a}},\end{aligned}$$

since $\sqrt{1-\frac{1}{4}a^2} = \frac{1}{2}\sqrt{4-a^2}$, and doubling the top and bottom of the fraction inside the inverse tangent. This is the required result.

The second "hence"

Here we take out $\sqrt{2}$:

$$\frac{1}{\sqrt{2}+\cos x} = \frac{1}{\sqrt{2}}\cdot\frac{1}{1-\left(-\frac{1}{\sqrt{2}}\right)\cos x},$$

so we use $a = -\frac{1}{\sqrt{2}}$ (which lies in $(-1,1)$) and $\theta = \frac{3}{4}\pi$ (which lies in $(0,\pi)$). Then $\sqrt{1-a^2} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$, and

$$\frac{1+a}{1-a} = \frac{1-\frac{1}{\sqrt{2}}}{1+\frac{1}{\sqrt{2}}} = \frac{\sqrt{2}-1}{\sqrt{2}+1} = (\sqrt{2}-1)^2,$$

on multiplying top and bottom by $\sqrt{2}-1$ and using $(\sqrt{2}+1)(\sqrt{2}-1)=1$. So the square root is $\sqrt{2}-1$, and the general formula gives

$$\int_0^{\frac{3}{4}\pi} \frac{1}{\sqrt{2}+\cos x}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\cdot\frac{2}{1/\sqrt{2}}\tan^{-1}\Big((\sqrt{2}-1)\tan\tfrac{3}{8}\pi\Big) = 2\tan^{-1}\Big((\sqrt{2}-1)\tan\tfrac{3}{8}\pi\Big).$$

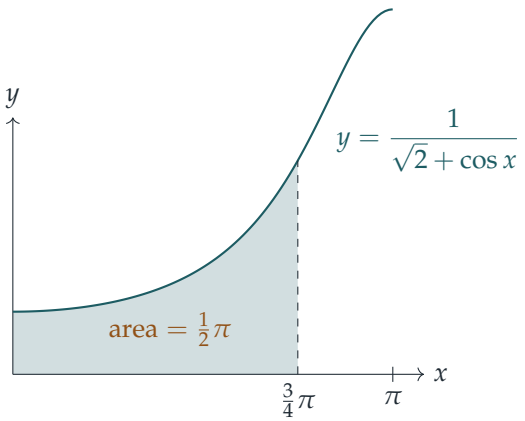
Ask yourself What is $\tan\frac{3}{8}\pi$?

Use the double-angle formula with $u=\tan\frac{1}{8}\pi>0$: $1=\tan\frac{1}{4}\pi=\dfrac{2u}{1-u^2}$, so $u^2+2u-1=0$ and $u=\sqrt{2}-1$ (the positive root). Since $\frac{3}{8}\pi=\frac{1}{2}\pi-\frac{1}{8}\pi$, we get $\tan\frac{3}{8}\pi=\cot\frac{1}{8}\pi=\dfrac{1}{\sqrt{2}-1}=\sqrt{2}+1$.

Hence $(\sqrt{2}-1)\tan\frac{3}{8}\pi=(\sqrt{2}-1)(\sqrt{2}+1)=1$, and

$$\int_0^{\frac{3}{4}\pi} \frac{1}{\sqrt{2}+\cos x}\,\mathrm{d}x = 2\tan^{-1}1 = 2\cdot\frac{\pi}{4}=\frac{\pi}{2},$$

as required.



Checks

- With $a=0$ the general formula gives $2\tan^{-1}(\tan\frac{1}{2}\theta)=\theta$, which is $\int_0^\theta 1\,\mathrm{d}x$. Similarly, $a=0$ in the first "hence" gives $\frac{2}{2}\tan^{-1}1=\frac{1}{4}\pi=\int_0^{\pi/2}\frac{1}{2}\,\mathrm{d}x$.
- Differentiating the boxed answer with respect to θ should give back $1/(1-a\cos\theta)$; it is a good exercise in the chain rule.
- Simpson's rule on the final integral gives 1.5707963…, matching $\frac{1}{2}\pi$; the general formula and the first "hence" also agree with Simpson's rule to within 10^{-10} for hundreds of random choices of a and θ.

Common slips

- Forgetting the factor $\frac{1}{2}$ (or $\frac{1}{\sqrt{2}}$) taken outside when rescaling the integrand to the form $1/(1 - a\cos x)$.
- Using $a = +\frac{1}{\sqrt{2}}$ in the last part: the sign in $\sqrt{2} + \cos x$ means a is negative.
- Losing track of k: the answer is $\frac{1}{k}\tan^{-1}(t/k)$, not $\tan^{-1}(t/k)$ or $\frac{1}{k}\tan^{-1}(kt)$.

Beyond STEP

The substitution $t = \tan\frac{1}{2}x$ (often called the Weierstrass substitution) works because it parametrises the unit circle by rational functions, so it reduces any rational function of $\sin x$ and $\cos x$ to a rational function of t, which can always be integrated by partial fractions. Integrals like $\int_0^{2\pi} \mathrm{d}x/(1 - a\cos x) = 2\pi/\sqrt{1 - a^2}$ are also standard first examples of contour integration in complex analysis, where the substitution $z = \mathrm{e}^{\mathrm{i}x}$ and the residue theorem do the work. For that viewpoint, see Needham, *Visual Complex Analysis*.

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